Tunis Business School



Department of Finance

FIN 460 – Dynamic Asset Pricing Theory

A Nowcasting Model for the Tunisian Dinar

Official Fixing versus Market Rate under a Managed Float

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Abstract

The Tunisian dinar trades under a managed float with capital controls. The Central Bank of Tunisia (BCT) publishes an official USD/TND fixing twice a day, while banks settle foreign currency at an interbank rate that is only reported with a one-day lag. We study two questions on daily data from April 2020 to April 2025. First, can the interbank rate be nowcast in real time from same-day movements in the major currency pairs (EUR/USD, GBP/USD, USD/JPY)? Second, how, and when, does the administered fixing diverge from the market rate?

We model the fixing log-return as a currency basket with weights estimated by ordinary least squares and by a Kalman filter, add a GARCH(1,1) volatility component selected by information criteria, and forecast the interbank rate as the fixing nowcast plus a separately modelled fixing-to-market spread. On the held-out test year the basket-plus-spread nowcast reaches a mean absolute percentage error of 0.17% and 63% directional accuracy on the interbank rate, ahead of a random-walk benchmark (0.20%, 53%); a neural-network residual correction trained strictly out of sample adds nothing. Consistent with the exchange-rate literature, the level metrics are flattering for a slowly-moving managed rate, so we treat directional accuracy as the informative measure.

The more distinctive result concerns the divergence itself. The interbank rate sits above the fixing on 62% of days, the spread widens to 0.12 dinar during the 2022–2023 foreign-exchange liquidity squeeze, and the 60-day correlation between fixing and interbank returns falls from about 0.8 to 0.35 over the same period before recovering. The official rate decouples from the market precisely when external pressure is greatest, which is the angle a free-floating treatment of the dinar would miss.

Contents

Abstract **1**

1 Introduction **3**

2 Data **3**

2.1 Series and sources 3

2.2 Why the two dollar series differ 4

2.3 Transformation and stationarity 4

3 Methodology **5**

3.1 The basket model 5

3.2 Estimating the weights 6

3.3 Volatility 6

3.4 From fixing to interbank: the spread 7

3.5 Residual correction 7

3.6 Nowcasting protocol and evaluation 7

4 Results **8**

4.1 Basket weights 8

4.2 Nowcasting the fixing and the interbank rate 8

4.3 The spread carries global information 9

4.4 Volatility 9

4.5 Official versus market: the divergence 10

4.6 How much does the basket lean on the euro? 11

5 Limitations and next steps **12**

6 Reproducibility **13**

7 Context and contribution **14**

License and disclaimer **14**

1 Introduction

Tunisia runs a managed float. The dinar is not pegged, but it is not freely floating either: the Central Bank of Tunisia (BCT) steers it within a framework of capital controls and publishes an official USD/TND fixing twice a day. Because Tunisia is a net importer with recurring external financing needs, the level of the dinar is a first-order macroeconomic variable, and the mechanics of how it is set matter as much as where it ends up.

Two prices coexist. The **official fixing** is the administered reference rate published by the BCT. The **interbank rate** is the price at which Tunisian banks actually exchange dollars, and it is the one that reflects domestic foreign-currency supply and demand. The two are related but not identical: the interbank rate is reported with a one-day lag, and it drifts away from the fixing whenever local liquidity tightens. The gap between them is therefore an observable symptom of foreign-exchange stress that a single official series would hide.

This lag creates a concrete problem. A desk that needs the interbank rate today cannot read it off a screen; it can only see yesterday's print and today's moves in the global majors. The first goal of this project is to close that gap with a *nowcast*: an estimate of today's interbank USD/TND built from information available in real time, namely the same-day returns of EUR/USD, GBP/USD and USD/JPY. We treat the dinar fixing as a currency basket, let the basket weights vary over time through a state-space model, and add a separate model for the fixing-to-interbank spread.

The second goal is to take the official-versus-market distinction seriously as an object of study rather than a nuisance. Most exchange-rate modelling concerns free-floating currencies, where a single market price is the only price. For a managed currency the wedge between the administered fixing and the market rate carries information about capital controls and external pressure, and quantifying it—its size, its persistence, and how it behaves during stress—is where this work departs from a textbook basket exercise.

We are deliberately modest about the models themselves. The basket regression, the Kalman filter and the GARCH volatility model are standard tools (Kalman, 1960; Bollerslev, 1986; Harvey, 1989), and Meese and Rogoff (1983) is the standing reminder that structural exchange-rate models rarely beat a random walk out of sample. We benchmark against that random walk throughout, report directional accuracy rather than leaning on level-fit statistics, and let the data say plainly where the model helps and where it does not.

2 Data

2.1 Series and sources

The study uses daily data from a single workbook with three sheets:

- **BCT fixing** — the official USD/TND fixing with bid, ask and mid quotes. The mid is our fixing series; the bid/ask spread is kept as a liquidity proxy.
- **Interbank rates** — the rate at which banks trade, quoted for the dollar and for roughly twenty other currencies against the dinar. We use the dollar leg as the interbank USD/TND.
- **Major pairs** — EUR/USD, GBP/USD and USD/JPY, the global majors that drive the basket.

The fixing and the majors run from April 2020 to April 2025; the interbank panel reaches back to 2004 but only overlaps the fixing from 2020. After aligning on common trading days and dropping incomplete rows we are left with **1,260 daily observations** spanning 2020-04-13 to 2025-04-07. The broader interbank panel is available for a wider basket but is not needed here: three majors are enough to span the dinar's daily variation, and adding thinly-quoted crosses would introduce more noise than signal.

2.2 Why the two dollar series differ

For a freely floating currency the official and market quotes would be the same number. For the dinar they are not, and the difference is the point. The fixing is an administered rate; the interbank rate clears the local market for foreign currency. When importers compete for scarce dollars—around debt-service dates or import surges—the interbank rate rises above the fixing and the spread widens. Keeping both series, rather than collapsing them into one "the dinar" price, is what lets us measure that pressure directly (Section 4.5).

2.3 Transformation and stationarity

All rates are converted to log-returns, $r_t = \log(P_t) - \log(P_{t-1})$, which removes the near-unit-root trend in the levels and gives roughly symmetric, comparable series. Augmented Dickey–Fuller tests reject a unit root in every return series at the 1% level (Table 1), so the returns are treated as stationary for the modelling that follows. The fixing-to-interbank spread, $s_t = \mathrm{IB}_t - \mathrm{Fixing}_t$, is kept in price (dinar) units because it is itself the quantity of interest.

Provenance and sharing. The series are terminal exports of BCT and market quotes collected for this coursework. The sample ends on 2025-04-07, the last date present in the workbook; the model cannot speak to anything after that without an updated feed. The file is kept in the repository so the pipeline reproduces exactly, but it should be treated as coursework data rather than a redistributable dataset.

Table 1: Augmented Dickey–Fuller tests on daily log-returns. The null of a unit root is rejected for all series ($p < 0.001$).

Series	ADF statistic	1%	5%	10%
EUR/USD return	−26.07	−3.44	−2.86	−2.57
GBP/USD return	−26.03	−3.44	−2.86	−2.57
USD/JPY return	−17.69	−3.44	−2.86	−2.57
Fixing return	−35.23	−3.44	−2.86	−2.57

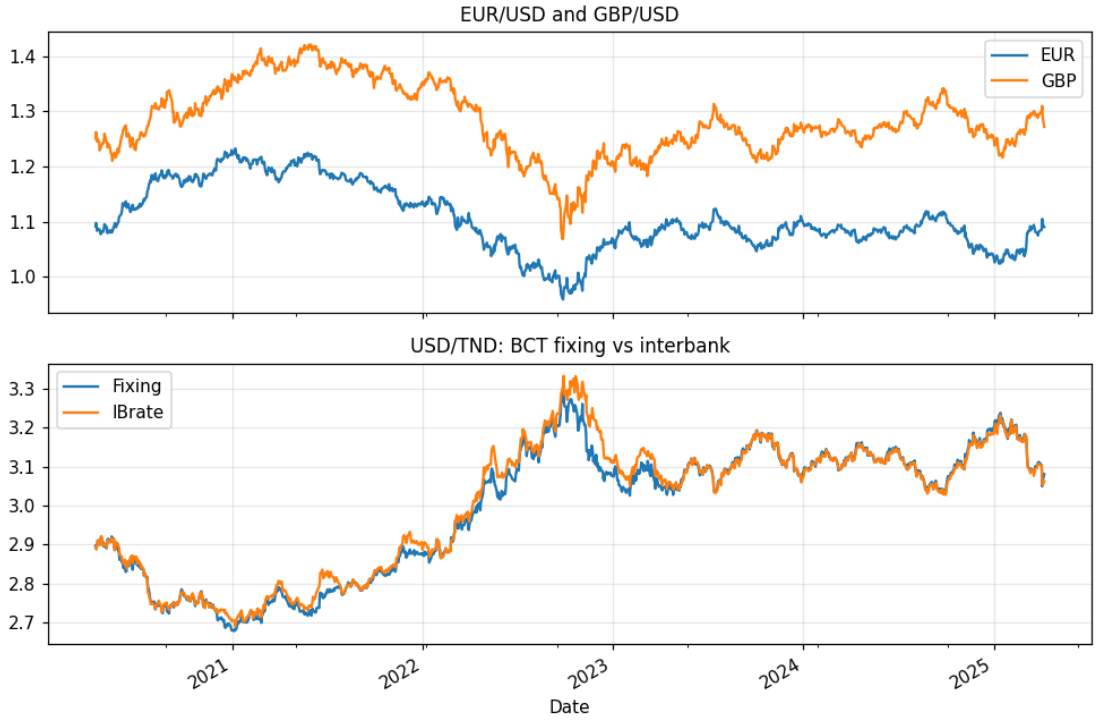


Figure 1: Top: the two majors used in the basket. Bottom: the BCT fixing and the interbank USD/TND move together but the interbank rate sits above the fixing, most visibly in 2022–2023.

3 Methodology

3.1 The basket model

We write the fixing log-return as a weighted combination of the major-pair log-returns,

$$\Delta \log(\text{Fixing}_t) = \mathbf{w}_t^\top \Delta \log(\mathbf{X}_t) + \varepsilon_t, \qquad \mathbf{X}_t = (\text{EUR/USD}_t,\ \text{GBP/USD}_t,\ \text{USD/JPY}_t), \tag{1}$$

and reconstruct the level by anchoring to the previous fixing, $\widehat{\text{Fixing}}_t = \text{Fixing}_{t-1} \exp(\Delta \widehat{\log \text{Fixing}}_t)$. The question is what to assume about the weights $\mathbf{w}_t$.

3.2 Estimating the weights

We tried four estimators, in increasing order of flexibility.

Constant weights (OLS). The natural benchmark fixes $\mathbf{w}_t = \mathbf{w}$ and estimates it by ordinary least squares on the training span. This is the model Equation (1) reduces to under constant weights, and it is the baseline every later model has to beat.

Online weights (SGD). To let the weights drift we updated an `SGDRegressor` one day at a time with `partial_fit`, feeding the standardised returns sequentially. In practice the recovered weights barely move: the online path stays close to the OLS values rather than tracking a genuinely time-varying relationship. Stochastic gradient updates on a single noisy observation per step do not, on their own, produce informative time variation.

Inverted-target weights (Random Forest). A second attempt inverted Equation (1) day by day—solving, in a least-squares sense, for the weights that reproduce the recent fixing moves—and trained a random forest to predict those weights from recent returns. The inverted targets turn out to be extremely unstable (a standard deviation near 0.7 across days, because three weights are being identified from a handful of noisy returns), and the forest simply regresses toward their average. Reconstructing the fixing from these weights gives nothing over the constant-weight benchmark, so the approach was dropped.

Random-walk weights (Kalman filter). The estimator we keep treats the weights as a latent state that follows a random walk and is updated by a Kalman filter (Kalman, 1960; Durbin and Koopman, 2012):

$$\text{state:} \quad \mathbf{w}_t = \mathbf{w}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(0, Q), \tag{2}$$

$$\text{observation:} \quad \Delta \log(\text{Fixing}_t) = \mathbf{w}_t^{\top} \Delta \log(\mathbf{X}_t) + v_t, \quad v_t \sim \mathcal{N}(0, R). \tag{3}$$

The filter produces one-step-ahead weight estimates using only past information, which is what we want for a nowcast. We carry a constant term in the state so the filter is not forced through the origin, and we choose the process and observation variances (Q, R) by a small grid search that minimises the one-step validation error rather than by maximum likelihood, which keeps the filter deliberately smooth. The resulting weights vary slowly, in line with the finding above that the basket relationship is close to constant.

3.3 Volatility

Daily fixing returns are heteroscedastic, so we fit a GARCH model (Engle, 1982; Bollerslev, 1986) to the fixing return and select the order by information criteria, comparing GARCH$(1,1)$, $(1,2)$, $(2,1)$ and $(2,2)$. The conditional volatility serves two purposes: it

is a description of how dinar risk evolves, and—fed in as a time-varying observation variance R_t—it lets the Kalman filter discount noisy, high-volatility days. We report both the selection and the effect of the volatility-weighted filter in the results.

3.4 From fixing to interbank: the spread

The interbank rate is the fixing plus the market spread, so we model it that way:

$$\widehat{\mathrm{IB}}_t = \widehat{\mathrm{Fixing}}_t + \widehat{s}_t, \qquad s_t = \mathrm{IB}_t - \mathrm{Fixing}_t. \tag{4}$$

This keeps the two sources of variation separate—global moves in the fixing, local pressure in the spread—instead of folding them into one regression. The spread is stationary and we nowcast it three ways: an AR(1), an ARIMAX(1,0,1) that adds the same-day major returns as exogenous regressors (Box and Jenkins, 1976), and a random forest on lagged spreads and recent spread volatility. Comparing them tells us whether the spread carries information beyond its own persistence.

3.5 Residual correction

Finally we ask whether anything systematic is left in the interbank residual, $\mathrm{IB}_t - \widehat{\mathrm{IB}}_t$. A small multilayer perceptron is trained on the residual's own lags and recent spread volatility. Crucially, it is fitted on the training span only and scored on the untouched test span; predicting the test residuals from a model that has also seen them would manufacture an improvement that does not exist out of sample. The honest test is whether the correction helps on data the network never trained on.

3.6 Nowcasting protocol and evaluation

We split the sample chronologically, 80% for training and the final 20% (the year to April 2025) for testing, and never let test data influence estimation. At each test day the nowcast uses the same-day major returns—observable in real time—to estimate the not-yet-published interbank rate. We report root mean squared error, mean absolute percentage error, correlation and directional accuracy, and we benchmark every model against a random walk that simply carries yesterday's rate forward.

One caution shapes how we read these numbers. A managed rate moves little day to day, so any sensible level nowcast will show a low MAPE and a correlation near one—the random walk already does. Those statistics mostly measure the persistence of the level, not skill. Directional accuracy, the share of days on which the predicted move has the right sign, is the metric that actually separates the models, and it is the one we lead with.

4 Results

All figures and numbers below come from the held-out test year, 2024-04-02 to 2025-04-07 (252 trading days), with models estimated only on the prior four years.

4.1 Basket weights

The OLS weights are −0.35 on EUR/USD, −0.04 on GBP/USD and 0.02 on USD/JPY. The negative euro loading is expected: the workbook quotes EUR/USD (dollars per euro), so a stronger euro coincides with a weaker dollar and a lower USD/TND fixing. The magnitude on the euro dwarfs the other two, which already signals that the dinar is, in practice, mostly a euro story—a point the sensitivity analysis makes precise. The Kalman filter's weights (Figure 2) drift only gently around these values, confirming that the basket relationship is close to constant over the sample.

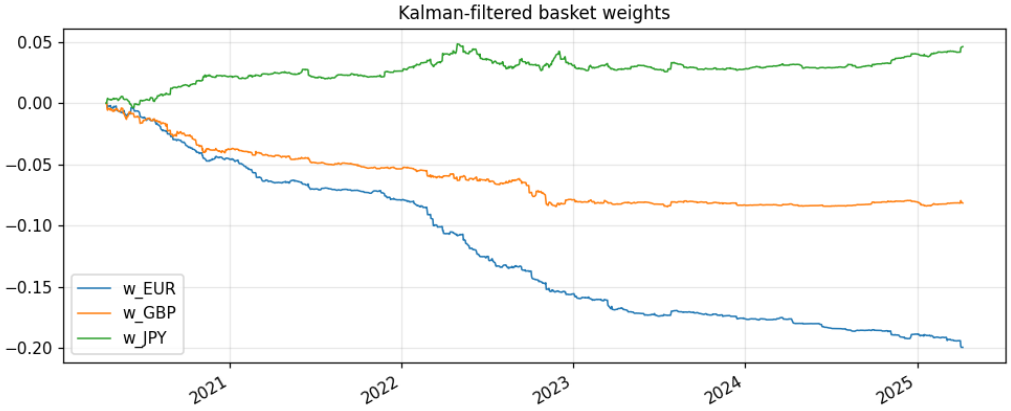


Figure 2: Kalman-filtered basket weights. The weights move slowly and stay near their OLS levels; the relationship is time-varying only mildly.

4.2 Nowcasting the fixing and the interbank rate

Table 2 collects the test-set performance. Two readings matter. On the headline level metrics every model looks excellent—correlations above 0.98, MAPE under a quarter of a percent—but so does the random walk, which is the warning from Section 3.7 made concrete. The separation shows up in directional accuracy. For the interbank rate the basket-plus-spread nowcast calls the daily direction correctly 63% of the time against the random walk's 53%, a real if modest edge. For the fixing the edge is smaller, and the constant-weight OLS is in fact as good as the Kalman filter on this sample—the dynamic weights buy interpretability, not accuracy.

The neural-network residual correction is the clearest negative result. In sample it can be made to fit the residuals almost perfectly, but trained on the training span and scored on the test span it leaves the interbank MAPE unchanged at 0.172% and slightly

Table 2: Out-of-sample nowcast performance. MAPE and correlation are flattering for a managed rate (the random walk scores well too); directional accuracy is the discriminating metric.

Target	Model	RMSE	MAPE (%)	Corr	DirAcc (%)
Fixing	OLS (constant weights)	0.0074	0.171	0.989	55.4
Fixing	Kalman (time-varying weights)	0.0087	0.203	0.985	53.8
Fixing	Random walk	0.0101	0.227	0.979	47.8
Interbank	Basket + ARIMAX spread	0.0075	0.172	0.988	**63.3**
Interbank	+ NN residual correction	0.0075	0.172	0.988	62.2
Interbank	Random walk	0.0090	0.205	0.983	53.4

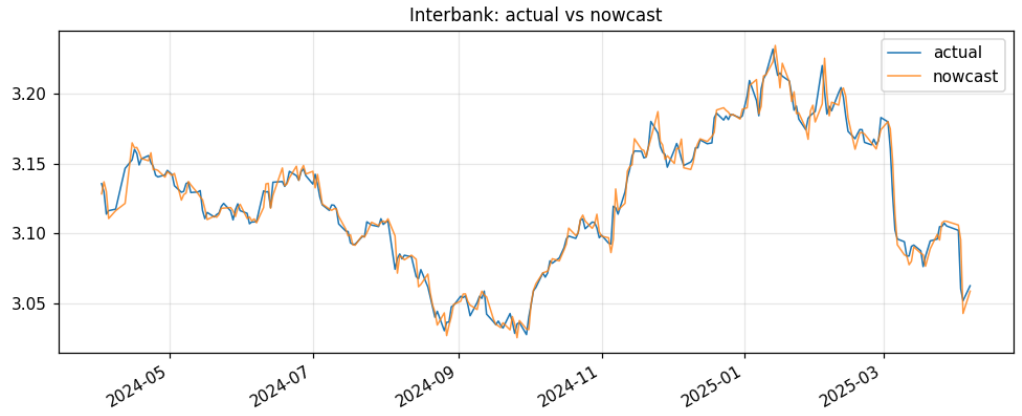


Figure 3: Interbank USD/TND: actual versus the basket-plus-spread nowcast over the test year.

lowers directional accuracy. The daily interbank residual, once the basket and the spread are accounted for, is essentially unpredictable. We report this rather than tune it away.

4.3 The spread carries global information

Decomposing the interbank rate into fixing plus spread pays off because the spread is not just inertia. Table 3 compares the three spread models. The AR(1) and the random forest, which rely on the spread's own lags, track it poorly (correlation below 0.1); the random forest puts 83% of its importance on a single lag and adds nothing for it. The ARIMAX, which feeds in the same-day major returns, more than doubles the correlation to 0.51. The spread moves with global FX, not only with yesterday's spread—which is why the same-day majors are useful for nowcasting the market rate and not just the fixing.

4.4 Volatility

Among the GARCH specifications, GARCH(1,1) minimises both AIC and BIC (Table 4); the higher-order models add parameters without improving the likelihood enough to

Table 3: Spread nowcast (interbank minus fixing), test set. MAPE is omitted because the spread crosses zero. The ARIMAX, with same-day major returns, is the only model that tracks the spread.

Model	RMSE	MAE	Corr
AR(1)	0.00653	0.00505	0.083
ARIMAX(1,0,1)	0.00451	0.00348	0.506
Random forest	0.00578	0.00448	0.040

justify them. Figure 4 shows the conditional volatility, which rises through the 2022–2023 stress and settles afterward. Using this series as a time-varying observation variance in the Kalman filter did not improve the fixing nowcast on this sample (it slightly worsened it), so we keep GARCH as a description of dinar risk rather than as a forecasting add-on.

Table 4: GARCH order selection on fixing log-returns. GARCH(1,1) is preferred by both criteria.

Specification	AIC	BIC
GARCH(1,1)	**6417.4**	**6437.9**
GARCH(1,2)	6418.4	6444.1
GARCH(2,1)	6419.2	6444.9
GARCH(2,2)	6420.4	6451.3

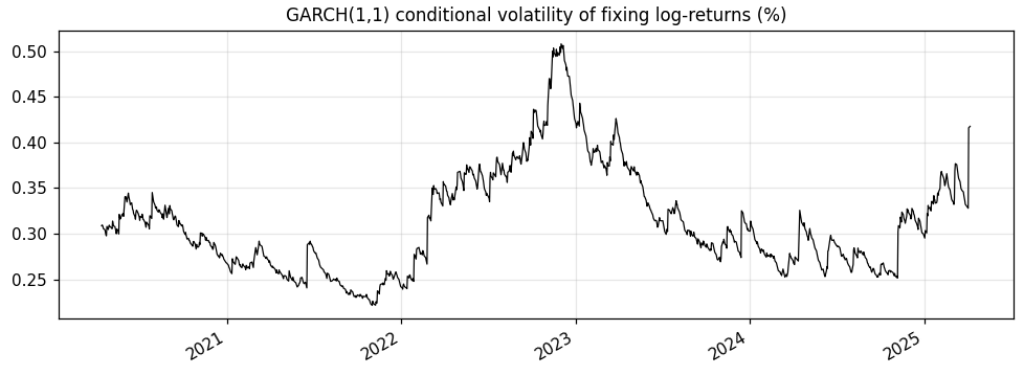


Figure 4: GARCH(1,1) conditional volatility of the fixing log-return. Risk clusters in the 2022–2023 liquidity squeeze.

4.5 Official versus market: the divergence

The distinctive result of the project is what happens between the two dollar series. Over the full sample the interbank rate sits above the fixing on 62% of days, by an average of 0.012 dinar (standard deviation 0.021), and the gap reaches 0.12 dinar at its widest. Figure 5 shows where that width concentrates: the spread is quiet through 2020–2021, blows out through 2022 and into early 2023, then compresses back toward zero. That

window coincides with Tunisia's acute foreign-currency shortage—import cover under strain and external financing in question—and the spread is reading exactly the local liquidity pressure that the official fixing is designed to smooth over.

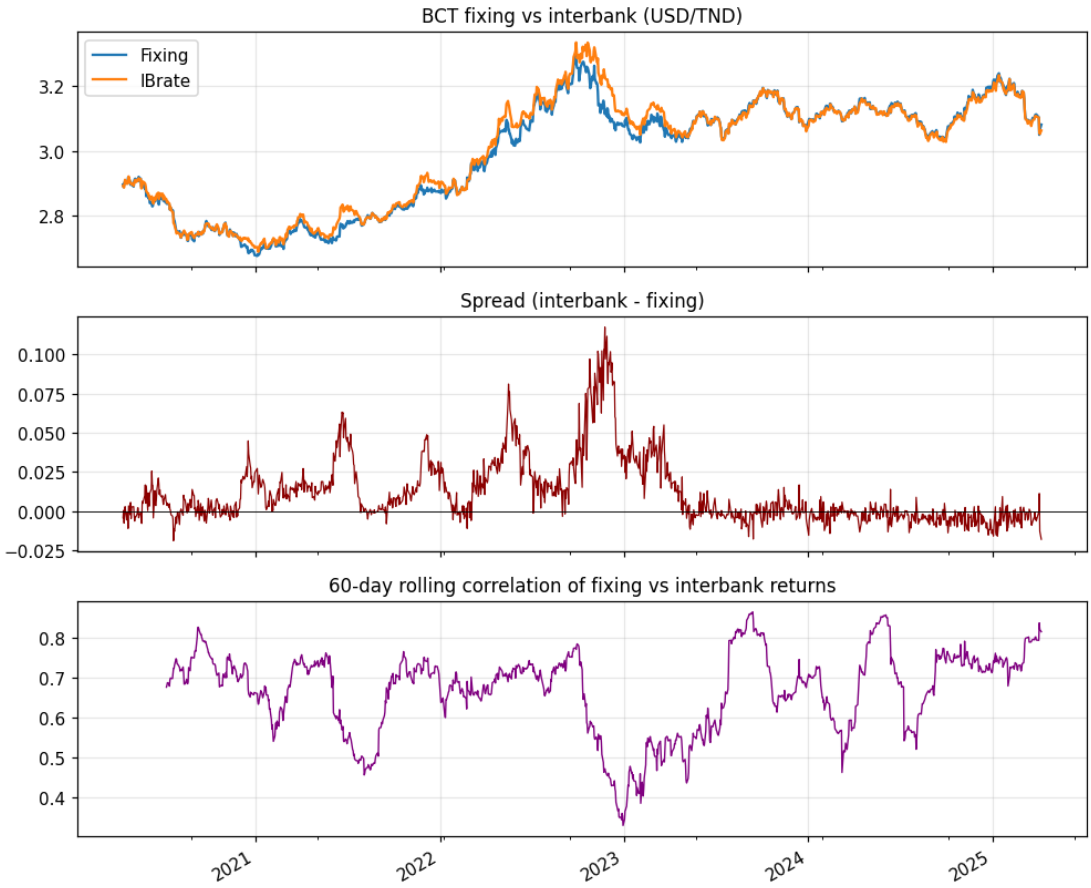


Figure 5: Official fixing versus interbank rate. Top: levels. Middle: the spread spikes to 0.12 dinar in 2022–2023. Bottom: the 60-day correlation of fixing and interbank returns falls from about 0.8 to 0.35 in the same window, then recovers.

The correlation panel is the sharper statement. In calm periods the fixing and the interbank rate move together, with a 60-day return correlation around 0.8. During the 2022–2023 squeeze that correlation collapses to about 0.35: the administered fixing and the market rate stop moving in step precisely when external pressure is greatest, then re-couple as conditions normalise. A free-floating treatment of the dinar, with one price, could not see this; it is visible only because we keep the official and market series apart. The fixing's own bid/ask spread, averaging about 10 basis points (Figure 6), tells a consistent if milder version of the same liquidity story.

4.6 How much does the basket lean on the euro?

Re-running the fixing nowcast with each weight bumped by ±50% (Figure 7) shows the valuation resting almost entirely on the euro leg. Changing the euro weight moves the test MAPE from its baseline of 0.17% up to 0.19%, while the same shocks to the sterling

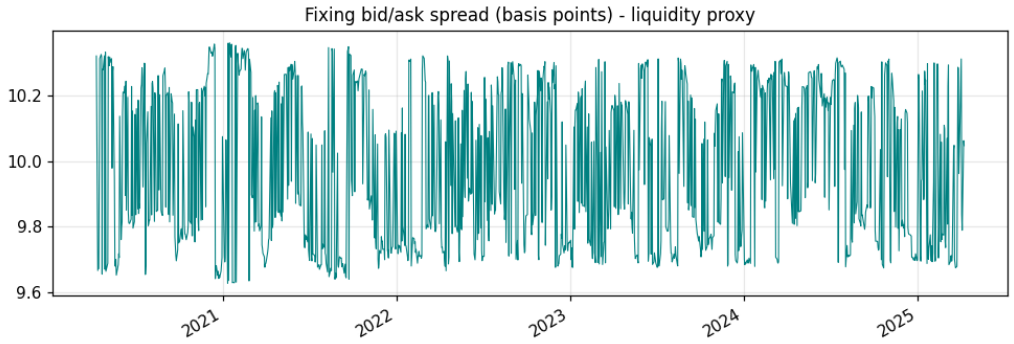


Figure 6: Bid/ask spread of the BCT fixing in basis points, a second liquidity proxy.

or yen weights barely register. The dinar fixing behaves, for nowcasting purposes, like a euro-dollar position with small corrections—consistent with the structure of Tunisia's trade and with the weights in Table 2.

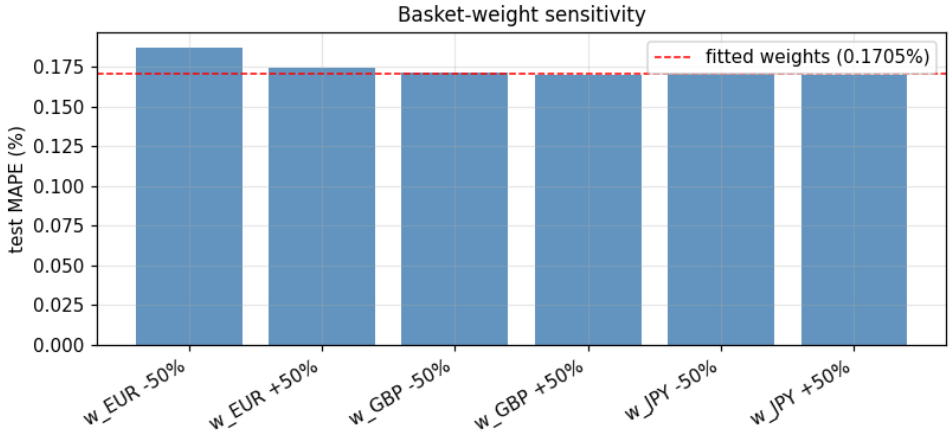


Figure 7: Test MAPE when each basket weight is bumped ±50%. Only the euro leg matters; the red line is the fitted-weight baseline.

5 Limitations and next steps

The benchmark is hard to beat for a reason. A managed rate barely moves from one day to the next, so the level statistics in Table 2 are generous to any model and the random walk is already close. The honest edge we find is in directional accuracy, and it is modest. This is the expected outcome for daily exchange-rate nowcasting (Meese and Rogoff, 1983), not a defect of the particular models, and it is why we avoid headline claims built on MAPE or correlation alone.

The fixing is administered, not market-clearing. The basket model assumes a stable statistical link between the dinar and the majors. The BCT can and does lean against that link, so the fixing is partly a policy choice rather than a market outcome.

The model captures the market-driven component of the fixing; it cannot anticipate a discretionary move by the central bank, and it should be read as describing the dinar's behaviour under the prevailing regime rather than predicting policy.

One sample, one regime split. The headline numbers come from a single 80/20 chronological split with the last year as the test set. We benchmark against a random walk and report directional accuracy to guard against reading too much into level fit, but a fuller study would use rolling-origin evaluation across several windows, including the 2022–2023 stress, to see how the edge holds up out of the calm.

The residual is unpredictable. The neural-network correction adds nothing out of sample. We take this at face value: once the basket and the spread are removed, the daily interbank residual carries no structure a model of its own lags can exploit. A larger or different network would not change that conclusion on this data, and we do not present one as if it did.

Data and timing. The sample ends on 2025-04-07, so the nowcast cannot be run live without an updated feed, and the same-day exogeneity we assume for the majors holds for a desk with real-time screens but not for an end-of-day reconstruction. The interbank "truth" is itself a reported series with its own lag.

Next steps. Three extensions follow naturally from the data already in hand. The interbank panel quotes the dinar against roughly twenty currencies back to 2004, which would support a wider basket and a longer history than the three-major sample used here. The spread model could be made regime-aware, switching between calm and stressed dynamics on the volatility signal the GARCH fit already provides. And the same fixing-versus-market decomposition could be applied to the dinar's euro leg, where the trade weights suggest the relationship is even tighter.

6 Reproducibility

The full pipeline is a set of Python modules under `src/` driven by a single entry point:

```
pip install -r requirements.txt
python main.py
```

This loads the workbook from `data/raw/`, builds the features, runs the OLS, Kalman, GARCH, spread and residual models, and writes every figure and metric in this report to `results/`. The numbers in Tables 2–4 are read directly from `results/metrics.json`; the figures are the files in `results/`. An optional real-time leg (`python main.py --live`) pulls spot majors from a data provider and applies the fitted weights; it requires an API

key supplied through the `ALPHAVANTAGE_KEY` environment variable and is not part of the reproducible run.

7 Context and contribution

This report was prepared for FIN 460, Dynamic Asset Pricing Theory, at Tunis Business School, under the supervision of Dr. Eymen Errais and Dr. Bechir Bouzid. It is a **group project** by Mustapha Aziz Laroussi, Salma Regaieg, Yassine Chakroun and Mohamed Mechri.

The work divided across data collection and cleaning, the modelling pipeline (basket weights, Kalman filter, volatility and spread models), the divergence analysis, and the writing. Mustapha Aziz Laroussi contributed to the modelling and analysis and to the preparation of this report.

License and disclaimer

The code is released under the MIT License. This is an academic exercise. Nothing here is investment advice, and the model is not intended for trading or policy decisions; it describes the behaviour of the Tunisian dinar over a fixed historical sample under the prevailing exchange-rate regime.

References

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

George E. P. Box and Gwilym M. Jenkins. *Time Series Analysis: Forecasting and Control*. Holden-Day, 1976.

James Durbin and Siem Jan Koopman. *Time Series Analysis by State Space Methods*. Oxford University Press, 2nd edition, 2012.

Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Andrew C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, 1989.

Rudolph Emil Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

Richard A. Meese and Kenneth Rogoff. Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, 14(1–2):3–24, 1983.